

Leveraging Large Language Models in Cyber Incident Detection and Response Systems

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Abstract—This paper analyzes current trends in the integration of Large Language Models (LLMs) into Security Operations Center (SOC) processes. The study summarizes findings on analyst interactions with GPT-4, examines the effectiveness of multi-model approaches for incident classification, and describes integration possibilities within open-source SIEM platforms. Particular attention is given to Retrieval-Augmented Generation (RAG) techniques and frameworks such as MoRSE, which aim to minimize model hallucinations and increase the accuracy of security threat analysis.

Index Terms—LLM, Cybersecurity, SIEM, SOC, RAG, Incident Detection, GPT-4, SOAR

I. INTRODUCTION

Recent research suggests that SOC analysts utilize LLMs primarily as on-demand cognitive aids focused on specific tasks rather than as continuous "collaborators." According to a study [1] that analyzed thousands of analyst-generated queries in an operational environment, LLMs are most frequently used for command explanation, scriptwriting, documentation improvement, and data analysis support.

II. ANALYSIS OF ANALYST INTERACTION WITH LLMs

The experiment described in [1] was conducted by capturing all queries sent to the GPT-4 model. A total of over 3,000 queries from 45 analysts were recorded over a 10-month period. A key finding was the growing trend in the frequency of LLM usage in the daily workflow of a SOC analyst, with 11 out of 45 analysts integrating GPT-4 into their everyday routine.

The research showed that only 4% of queries involved a request to directly evaluate an alert as malicious or benign, confirming the use of LLMs as assistive rather than decision-making tools. Furthermore, 75% of interactions consisted of only 2 to 3 questions, mostly involving iterative text refinement for incident reporting or explanations of functions and logs. The authors highlight the potential for integrating LLM tools directly into SIEM systems to reduce cognitive load by eliminating context switching.

III. MULTI-MODEL APPROACHES AND CLASSIFICATION

In [2], the authors explored a multi-model approach to classify security incidents into two categories: interesting and uninteresting. The highest success rate was achieved using GPT-4, reaching a precision of 94% and a recall of 92%.

Implementation of this approach led to a 40% reduction in incident processing time and a 60% decrease in cognitive load due to fewer manually handled alerts.

The experimental framework proposed by the researchers consists of a modular architecture featuring an Alert Generation Module, an LLM Agent, and a Reporting Module. This system was integrated into an environment utilizing the Wazuh SIEM platform, where CALDERA was employed for adversary emulation to generate realistic attack scenarios. This setup allowed for a comparative benchmark between five distinct models: GPT-4, GPT-3.5, LLaMA 3, Mixtral 8x22B, and OpenHermes 2.5 Mistral 7B. The inclusion of open-source models like LLaMA and Mixtral was intended to address critical privacy risks and data sovereignty concerns associated with processing sensitive SOC data through external APIs.

Despite the strong performance metrics of the proprietary models, the study highlights a significant operational hurdle: a hallucination rate of approximately 5% during the analysis of security events. While GPT-4 remained the top performer with an F1-score of 93%, the open-source alternatives demonstrated varying degrees of efficacy, trailing behind in complex classification tasks. This performance gap underscores the challenge of balancing model accessibility and privacy with the high accuracy required for autonomous alert triage.

To address these limitations, the authors advocate for a hybrid "co-pilot" strategy where LLMs act alongside human analysts. They propose further research into prompt optimization, model transparency, and bias detection to improve the reliability of the system. While the current iteration of the framework focuses on weighted outputs and scoring, the authors suggest that future developments should integrate continuous learning from large-scale alert training sets to mitigate the issues of redundancy and correlation between the models in the ensemble.

Despite these benefits, the study has limitations. The reported accuracy figures pertain to individual models, while the success rate of the multi-model voting itself is not clearly specified. Another open point is redundancy; if GPT-4 and GPT-3.5 are highly correlated, the third model's output may be consistently overruled. Furthermore, the nature of the "tuning" mentioned is unclear—whether it involved full fine-tuning or merely adjusting weights for final scoring—and a RAG-based approach was not utilized.

IV. INTEGRATION INTO OPEN-SOURCE SIEM

The authors in [3] present the Security Event Response Copilot (SERC) system, built on the open-source Wazuh SIEM. The system adds two components: a RAG-based module for context-aware incident retrieval and an LLM module for generating recommendations. The RAG system draws information from MITRE ATT&CK, the NIST Cybersecurity Framework (CSF), and an incident response knowledge base.

The architecture of SERC is designed to bridge the gap between strategic risk management and tactical technical execution. By utilizing three distinct vectorized data collections—the NIST CSF 2.0 for high-level lifecycle management, the MITRE ATT&CK framework for granular adversary behavior mapping, and the SOCFortress incident response plan for procedural ground truth—the system provides a multi-layered defense. This integration enables the LLM to deliver recommendations that are not only technically accurate in identifying tactics and techniques but also aligned with established organizational security standards.

The efficacy of the framework was evaluated through an end-to-end simulation using the Atomic Red Team tool to replicate six critical threat categories, including ransomware, malware, and phishing. To assess the quality of the generated responses, the researchers employed multi-faceted metrics such as cosine similarity for semantic alignment and BERTScore for contextual accuracy. This methodology allowed for a rigorous comparison between the proprietary OpenAI models and the open-source Pixtral model, focusing on their ability to minimize hallucinations—a persistent challenge when using LLMs in the SOC space.

The study also compares the performance of Pixtral and OpenAI models. Pixtral demonstrated high stability and consistency, while OpenAI showed greater adaptability but higher performance variability. Specifically, Pixtral consistently achieved higher precision and F1-scores (ranging between 85.4% and 87.2%) across diverse attack scenarios, reflecting its reliability in capturing relevant information without significant fluctuations. In contrast, while the OpenAI model peaked at higher semantic alignment (75.81% cosine similarity in certain cases), it exhibited more pronounced performance dips, suggesting that Pixtral may be a more stable candidate for consistent automated triage in open-source SIEM environments.

V. ADVANCED RAG FRAMEWORKS

Reference [4] addresses the exponential growth of unstructured cybersecurity information and LLM hallucinations. The proposed MoRSE framework is an open-source system integrating dual RAG modules in a cascade: a structured RAG for pre-processed data and an unstructured RAG for raw text. By drawing from CVE repositories, MITRE, and ExploitDB, MoRSE achieved a 15% improvement in general question accuracy and a 50% increase in accuracy for CVE-related incidents compared to GPT-4 without RAG.

The technical distinction of MoRSE (Mixture of RAGs Security Experts) lies in its parallel retriever architecture,

which was inspired by the Mixture of Experts (MoE) paradigm to handle multidimensional cybersecurity contexts. The framework operates in two distinct phases: an Information Retrieval phase, where specialized retrievers collect semantically related data in varied formats, followed by an Answer Generation phase driven by an LLM. Unlike traditional models that rely on static parametric knowledge, MoRSE utilizes a non-parametric knowledge base that allows for real-time updates without the need for resource-intensive retraining. This capability is particularly effective for complex "Multi-Hop" reasoning tasks, where the system must correlate disparate pieces of intelligence to identify root causes or predict subsequent stages of a cyberattack.

VI. AUTONOMOUS AGENTS FOR CYBER THREAT INTELLIGENCE

The automation of Cyber Threat Intelligence (CTI) analysis represents a critical frontier in reducing the manual workload of SOC analysts. As explored in [5] and [6], security teams traditionally rely on unstructured reports from sources such as CrowdStrike or MITRE ATT&CK to manually derive correlation rules for SIEM systems. This process is inherently slow and prone to human error. To address this, recent research proposes an autonomous AI agent capable of parsing natural language reports to identify Indicators of Compromise (IOCs) and automatically generating valid regular expressions (RegEx) for direct SIEM integration without human supervision.

The proposed system utilizes a graph-based approach to visualize relationships between IOCs, such as connecting specific file hashes to the domains and IP addresses involved in a singular attack campaign. However, achieving full autonomy requires overcoming four primary challenges: ensuring factual accuracy to filter hallucinations, translating complex natural language into syntactically correct RegEx, performing automated verification of these patterns, and maintaining the structural integrity of IOC relationship graphs.

Experimental results underscore the potential of this approach; in an analysis of over 50 CTI reports, the model identified approximately 2,900 IOCs. After autonomous filtering, 2,300 valid indicators remained, from which 2,200 RegEx patterns were successfully generated. Remarkably, the system missed only 3% of valid IOCs compared to manual ground truth. This shift toward self-learning cybersecurity automation marks a significant milestone in delegating repetitive SOC tasks to autonomous agents, thereby allowing human analysts to focus on higher-level strategic defense.

VII. LLMs IN DIGITAL FORENSICS AND TIMELINE ANALYSIS

Beyond real-time detection and threat intelligence, post-incident investigation—specifically Digital Forensics and Incident Response (DFIR)—presents a significant bottleneck for security teams. Constructing an accurate timeline of a cyber incident is a complex, labor-intensive task that requires parsing thousands of heterogeneous security logs to identify

correlations. The authors in [7] address this challenge by introducing GenDFIR, an innovative framework that leverages Retrieval-Augmented Generation (RAG) combined with Large Language Models to automate cyber incident timeline analysis.

By ingesting raw system events (such as Windows Security event logs detailing failed logons, credential validation errors, and audit log clearing) into a dynamic vector knowledge base, GenDFIR enables an LLM to accurately piece together the sequence of adversary actions. This methodology overcomes the traditional limitations of manual log triage, significantly accelerating the reconstruction of attack paths while mitigating human error and analyst fatigue. The RAG architecture is crucial in this context, as it anchors the LLM's outputs strictly to the ingested forensic artifacts, thereby preventing hallucinations and maintaining the evidentiary integrity of the analysis.

Evaluations of the GenDFIR framework demonstrated its capability to successfully reconstruct complex scenarios, such as unauthorized access chains, by maintaining strict chronological coherence and contextual relevance. By automating the extraction and chronological mapping of crucial artifacts, the system produces high-fidelity timeline reports that are directly applicable for formal incident reporting, root-cause analysis, and the formulation of subsequent defensive strategies.

VIII. PREDICTIVE THREAT INTELLIGENCE MODELS

Reference [8] investigates the integration of LLMs to overcome the limitations of static, signature-based detection systems that frequently struggle with zero-day exploits and polymorphic threats. The authors introduce the Predictive Threat Intelligence Model (PTIM), a framework trained on approximately 100 million security-related data points, including network logs and public threat databases. By utilizing the bidirectional context awareness inherent in transformer architectures, PTIM can recognize linguistic and non-linguistic patterns indicative of malicious intent that traditional models often overlook.

Empirical validation shows that PTIM achieves a detection accuracy exceeding 95% with a substantial reduction in false positives compared to rule-based systems. A key technical advantage of this framework is its operational efficiency; it performs significantly faster than standard LLM architectures, allowing for the near-instantaneous identification of ransomware attacks. This capability enables organizations to transition from a reactive posture to a proactive defense strategy, identifying and mitigating threats before they can cause widespread system compromise.

IX. LLM-BASED AGENTS FOR TIER 1 TRIAGE AUTOMATION

While previous research emphasizes broad intelligence gathering, [9] provides a specific evaluation of LLM agents within the critical triage process of Tier 1 SOC analysts. The study proposes a structured framework where an LLM agent serves as an intermediary between a SIEM platform (Wazuh) and the human analyst. By testing five prominent models—including

OpenAI's GPT-4o, Meta's Llama 3, and Mixtral 8x22B—the research investigates the agents' ability to autonomously classify alerts as either "interesting" or "not-interesting" based on raw log data. This classification logic is intended to filter out the high volume of "noise" generated by normal user operations, allowing analysts to focus on high-fidelity threats identified through adversary emulation tools like CALDERA.

Experimental findings indicate that LLM agents significantly enhance the speed of initial triage and reduce the cumulative cognitive load on security staff by providing natural language explanations for each alert classification. Despite these gains, the research highlights critical limitations such as the propensity for occasional hallucinations and the inherent privacy risks of processing sensitive log data through external models. Consequently, the study concludes that while LLM agents are highly effective for routine automation, they are best deployed in a "co-pilot" capacity. This hybrid approach ensures that human expertise remains in the loop to verify LLM reasoning, thereby maximizing operational efficiency while maintaining the rigorous security standards required in a modern SOC environment.

X. OPTIMIZATION OF SIEM DETECTION RULE SETS

While the automated generation of rules significantly expands detection coverage, it often introduces a secondary challenge: the proliferation of redundant or overlapping rules. Such overlaps increase computational overhead, response latency, and, most critically, alert fatigue for SOC analysts. Reference [10] introduces RuleGenie, a novel LLM-aided recommender system designed to optimize enterprise-level SIEM rule sets. The authors argue that manual optimization is increasingly unsustainable due to the sheer scale of modern rule bases, which can contain thousands of platform-specific queries (e.g., KQL, Sigma, or SPL) with complex logic.

The RuleGenie framework leverages the multi-head attention capabilities of transformer models to generate high-dimensional embeddings of SIEM rules. By applying a similarity matching algorithm to these embeddings, the system identifies the top-k most similar rules within a repository. Furthermore, RuleGenie utilizes Chain-of-Thought (CoT) reasoning to analyze the underlying logic of flagged rules, allowing it to recommend consolidations or deletions of redundant logic. This optimization process ensures that the SIEM maintains a lean and efficient detection posture, reducing the signal-to-noise ratio and ensuring that the most relevant alerts are prioritized for human investigation.

XI. AUTOMATED MAPPING OF SIEM RULES TO ADVERSARIAL TECHNIQUES

While the generation and optimization of SIEM rules are vital for visibility, the effectiveness of a security posture is often measured by its alignment with standardized frameworks like MITRE ATT&CK. As explored in [11], manual mapping of structured detection rules—such as those written in Splunk's Search Processing Language (SPL)—to specific Tactics, Techniques, and Procedures (TTPs) is a labor-intensive process

prone to subjective error. To resolve this, the authors propose Rule-ATT&CK Mapper (RAM), a multi-stage pipeline that utilizes Large Language Models to autonomously categorize structured rules. RAM is unique in its "prompt chaining" approach, which first translates technical rule syntax into natural language descriptions enriched by external contextual information (e.g., IoC significance) before performing the final technique recommendation.

Experimental evaluations using the Splunk Security Content dataset demonstrate that RAM significantly improves mapping accuracy without the need for model fine-tuning. By testing multiple models, including GPT-4-Turbo and IBM Granite, the study found that incorporating a web-search agent to retrieve supplementary contextual knowledge about Indicators of Compromise (IoCs) is essential for overcoming the limitations of an LLM's implicit domain knowledge. The framework achieved an F1-score of 0.62 and a recall of 0.75, highlighting its capability to assist SOC teams in maintaining an up-to-date and accurate coverage map of the ATT&CK matrix. This automation allows security engineers to rapidly validate whether their detection rule base provides sufficient protection against specific adversarial strategies.

XII. MULTI-AGENT MODULAR ARCHITECTURES FOR CORRELATED THREAT DETECTION

As cyberattacks become increasingly multi-vector and stealthy, isolated detection systems often fail to correlate related events across different layers of an organization's infrastructure. To address this, reference [12] proposes a modular multi-agent architecture that integrates domain-specific cybersecurity tools with the semantic reasoning capabilities of Large Language Models. The framework consists of three specialized autonomous agents—focused on email verification, server log analysis, and IP range scanning—which independently process data using tailored detection pipelines. These agents translate raw technical outputs into structured risk signals, which are then fed into a centralized contextual recommendation system.

The core innovation of this architecture lies in its ability to synthesize disparate proofs from individual agents into a coherent attack narrative. By cross-analyzing signals (e.g., correlating a suspicious phishing email with subsequent lateral movement in server logs), the system can detect complex threat patterns that evade standalone sensors. Experimental results on benchmark datasets, including CIC-IDS 2017 and SpamAssassin, demonstrate a high detection accuracy of 93.6% and a 41.3% reduction in false positives compared to traditional rule-based methods. Furthermore, the use of LLM-based Chain-of-Thought (CoT) reporting provides explicable insights for security analysts, significantly reducing triage time and increasing confidence in the system's automated recommendations.

XIII. OUR WORK - MIRA

XIV. CONCLUSION

The integration of LLMs into SOC processes represents a significant shift in the efficiency of cyber threat analysis. Future research should focus on deeper RAG integration and

the reduction of analyst cognitive load by minimizing context switching between LLM interfaces and primary security tools.

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